COMPSCI5079(M)
Cryptography & Secure Development

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Lecture 3:

Symmetric-key Cryptography
Block Ciphers

Lecture 3:

Overview

Outline

1. Feistel ciphers
   - A family of single-key, block substitution ciphers

2. Data Encryption Standard (DES)
   - 64-bit Block cipher, 56-bit key length
   - Insecure for modern applications
   - Still widely used

3. Advanced Encryption Standard (AES)
   - 128-bit Block cipher, different key lengths (128, 192, 256)
   - Replacement for DES

Feistel Ciphers: Summary

- The Feistel family of algorithms were designed to be fast.
  - The block size was two computer words. All the calculations were done in a single computer word of 32 bits (half of the block).
  - There were several rounds with sub-keys generated from the master key.
  - In each round:
    - A substitution is performed on the left half L_i
    - A permutation is performed by swapping the two halves.
  - Decryption used the same algorithm but with the subkeys in the reverse order.

Data Encryption Standard

- DES stands for Data Encryption Standard.
- It is now an older algorithm but still widely used.
  - There is a lot of legacy hardware and software.
- It is a Feistel block cipher, working with 64-bit data blocks and 56-bit keys.

DES Origins

- The NSA (US National Security Agency) modified the Lucifer cipher slightly and it was adopted in 1977.
  - They reduced the key length to 56 bits.
  - It uses elementary operations so that it is fast and easy to implement in silicon.

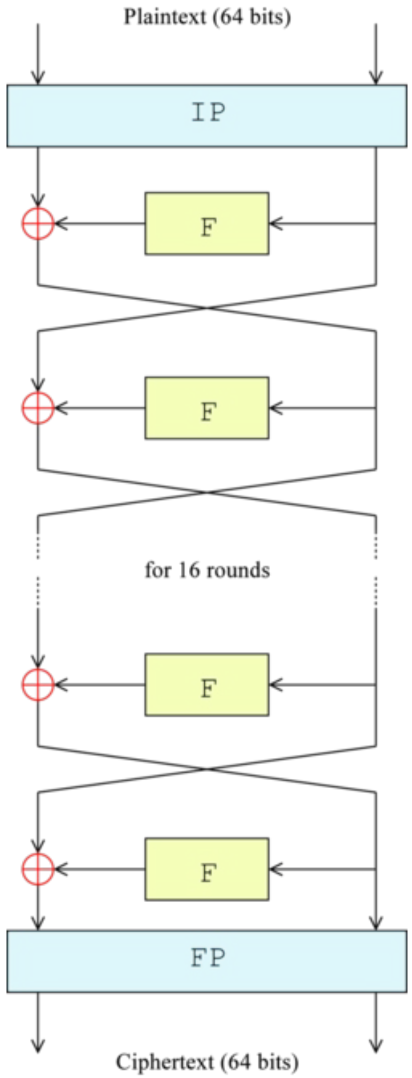
- The NSA did not publish an analysis of its security, encouraging some to think they had inserted a backdoor.
  - No back door has been found but NSA later said that they had changed the algorithm to protect against differential cryptography.
  - Rival algorithms, such as IDEAL (also a Feistel algorithm) in Europe, were found to be vulnerable to differential cryptography.

DES Overview

- DES is based on the Feistel cipher
- 16 rounds as in Feistel structure

$$L_{i+1} = R_i$$
$$R_{i+1} = L_i \oplus F(R_i, K_i)$$

- Two additional rounds at the beginning Initial Permutation (IP) and at the end Final Permutation (FP)
- We will go into detail about IP, FP, F function, and sub-keys generation from a master key



https://en.wikipedia.org/wiki/Data_Encryption_Standard

Simplified DES

- To understand how DES works, we study a toy system Simplified-DES (S-DES) developed by Edward Schaefer of the University of Santa Clara.

- S-DES encrypts 8-bit blocks of data using a 10-bit key.
- There are two substitution rounds, each with its own sub-key.
- Left and right halves are swapped between them.

- Let us generate a sample data block and sample key to work through the S-DES algorithm.

- Data = 00111110
- Key = 1011000110

The Initial Permutation

- The initial permutation (IP) operates on the 8-bit data blocks.
- If the data bits are numbered 0…7 then the Initial Permutation is 15203746.

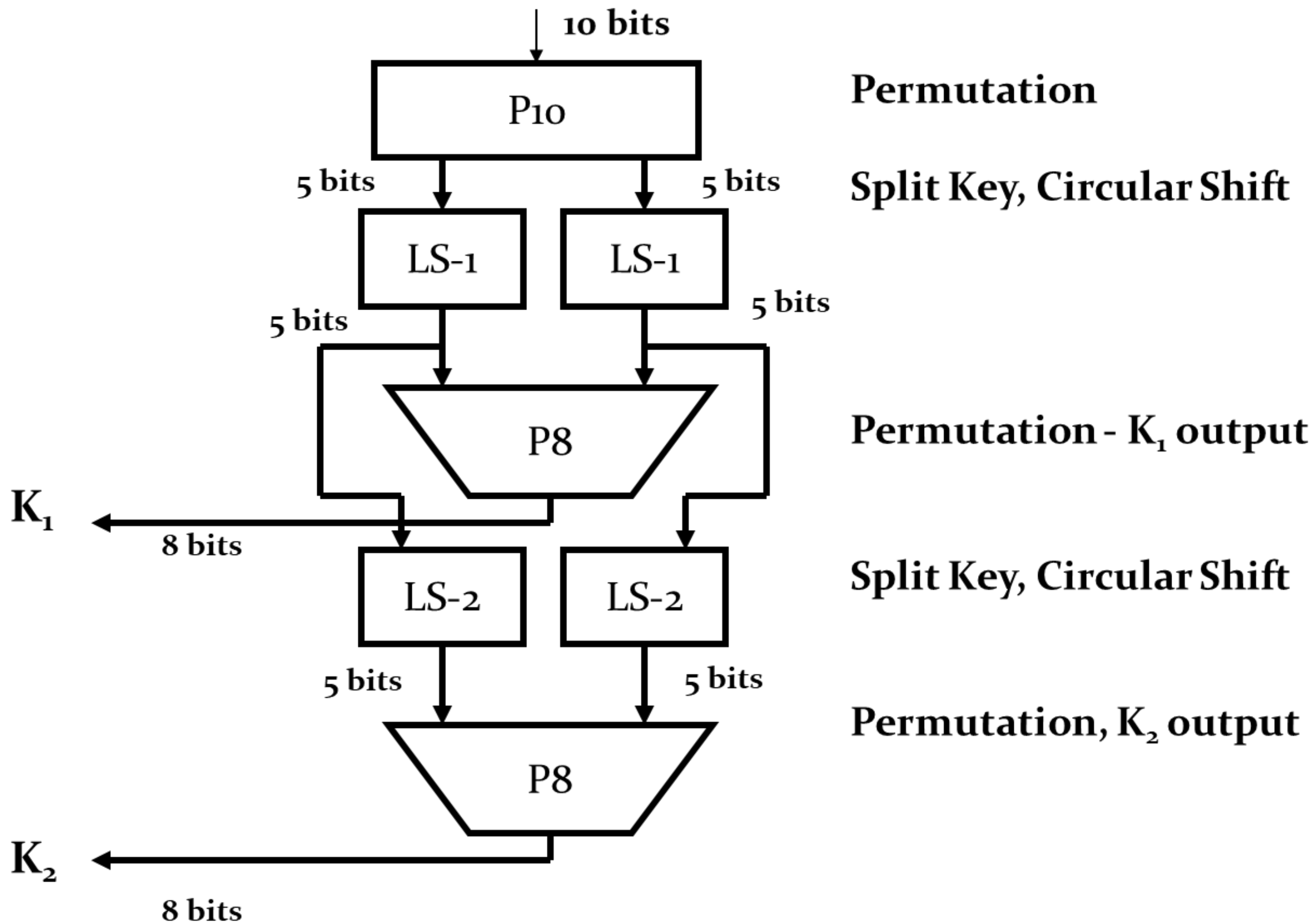
IP8							
1	5	2	0	3	7	4	6

- Permuted data is: 00111110 → 01101011

- The inverse permutation (Final Permutation) applied at the end of the process is 30246175.

FP8							
3	0	2	4	6	1	7	5

Sub-key Generation



Sub-keys Generation

Two 8-bit sub-keys are generated from the 10-bit master key.

Step 1: The bits of the original key are permuted based on P10 permutation table:

P10									
2	4	1	6	3	9	0	8	7	5

Key = **1011000110 → 1000101110**

Step 2: The left 5 bits and the right 5 bits are both rotated left 1 bit.

10001 01110 → 00011 11100

Step3: The first 8-bit sub key is formed from the following matching 8-bit table:

P8							
5	2	6	3	7	4	9	8

We then have: K_1 = **10111100**

Sub-keys Generation

Step 4: The left and right 5 bits from step 2 are both rotated left 2 bits

00011 11100 → 01100 10011

Step 5: The second 8-bit sub key is formed from the same bits as step 3 (using P8 permutation table again)

P8							
5	2	6	3	7	4	9	8

K_2 = ???

F function in details

- The most complex component in S-DES is the function F
- Defined by steps as shown in the diagram:

Step 1: Expansion of 4 data bits (right half of block) to 8 bits following the E/P table 30121230

Permuted data:

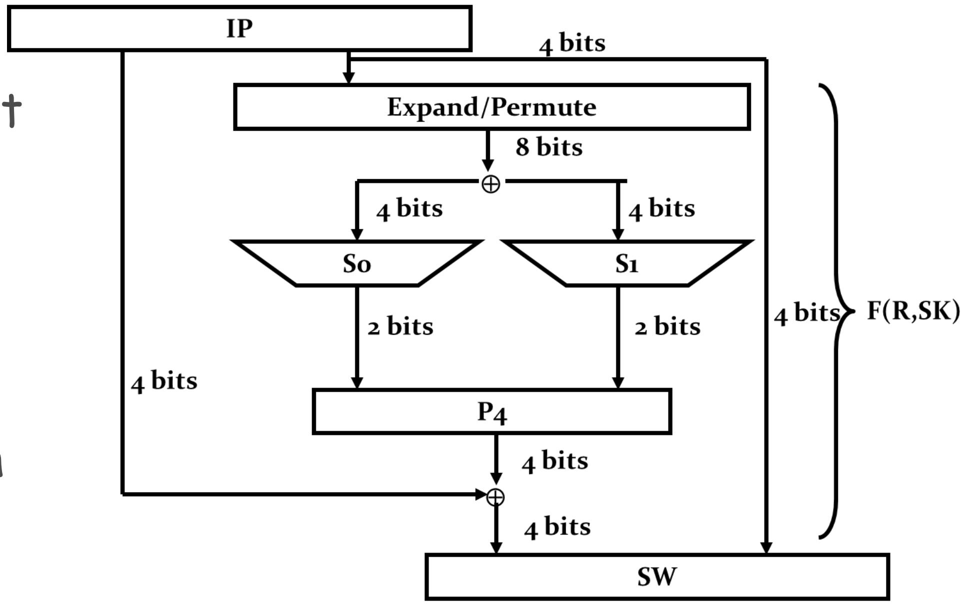
0110 1011 → 11010111

Step 2: These 8 bits are XOR'ed with the 8 bits of the first key K_1.

11010111

<u>10111100</u> (Key)

Output: 01101011



Simplified DES - Function F(R,SK)

F function in details

Step 3: The resulting 8 bits (numbered *0 1 2.. 7*) are split again into 4 separate 2-bit numbers called (row_1, col_1), (row_2, col_2) as following rule:

row_1 = bits: 0,3; col_1 = bits: 1,2
row_2 = bits: 4,7; col_2 = bits: 5,6
01101011 ➔ (00, 11) , (11, 01).

In decimal, $row_1 = 0$, $col_1 = 3$, $row_2 = 3$, $col_2 = 1$.

Step 4: (row_1,col_1) and (row_2,col_2) form the row and column indices of two 4x4 tables called Substitution-boxes (S-boxes)

- An S-box is a matrix, indexed by the row and column.
- Each S-box produces 2 bits of output.

S1(0, 3) = 10, S2(3, 1) = 01

So, output = 1001

S1				S2			
1	0	3	2	0	1	2	3
3	2	1	0	2	0	1	3
0	2	1	3	3	0	1	2
3	1	0	2	2	1	0	3

Substitution Boxes (S-Boxes) lookup table in decimal

F function in details

Step 5: The resulting 4-bit number, the output from the two S-boxes undergoes another permutation with P4 table 1320 which is the output of the function F

P4			
1	3	2	0

The algorithm continues:

Step 6: The Switch function (SW) interchanges the left and right 4 bits so that the second instance of f K operates on a different 4 bits.

Step 7: In this second instance, the E/P, S0, S1, and P4 functions are the same. The key input is K2.

Step 8: Finally apply inverse permutation (i.e., Final Permutation) to get the ciphertext.

Relationship with the Real DES

- DES has the same structure as S-DES, but with more steps.

- There are 16 F_K steps, each with a 48-bit sub-key generated from the 56-bit actual key.
- The function F operates on 32-bit halves of the data.
  - The data is expanded to 48 bits and XOR'ed with the sub-key.
  - These 48 bits are split into 8 chunks, each 6 bits long.
  - Each 6-bit chunk is treated as row (2 bits) and column (4 bits)
  - They each index an S-box.
- Internally it has 8 S-Boxes, each 4 x 16.
  - Each S-Box produces a 4-bit number.
  - The 8 S-Boxes produce a 32-bit value.

Results from more steps in DES

- The following table lists the number of bits that have changed after each round of DES with
  - two very similar plaintext blocks (diffusion).
  - two very similar keys (confusion).

- Clearly, diffusion and confusion are quite effective.

- Permutations on their own do not affect confusion or diffusion.
  - 1-bit change in the origin only changes one bit in the destination.

16 steps: Number of Different Bits in DES

Round	Confusion	Diffusion
1	6	2
2	21	14
3	35	28
4	39	32
…	…	…
10	44	38
11	32	31
12	30	33
13	30	28
14	29	34
15	29	34
16	34	35

Design of the S-Boxes

- The design principles were published in 1992, answering questions that NSA had introduced a trap door.
  - There was no trap door.
- The design made DES resistant to differential cryptanalysis, which NSA had known about but kept secret.
- Other Feistel ciphers were vulnerable to differential analysis.
- Differential Cryptanalysis uses two very similar chosen plaintext messages to uncover details of the encryption algorithm.
- Linear cryptanalysis is a similar attack that relies on two similar known plaintext messages.
  - DES is also resistant to it.

Breaking DES

- The short key length of 56 bits makes DES vulnerable to a brute force attack, where all keys are tried.
- On 29th Jan 1997, the RSA organisation offered a prize of $10,000 to the first person to find the key to some ciphertext when they were given 3 blocks of plaintext (a known plaintext attack).
  - Rocke Versur claimed the prize on 25th May 1997.

- In 1976 Hellman and Diffie estimated that it would cost $20M to build a special-purpose machine to crack DES in 1 day.
- In 1998 The Electric Frontier Foundation built a DES cracker for $250,000. It could search all possible keys in 9 days.
- DES lasted 20 years before it became easy to crack.

Double DES

- Double DES uses two different keys to encrypt twice.
  - $C = E_{K2}(E_{K1}(P))$

- It is vulnerable to a known plaintext (assume P and C are known): *meet in the middle attack.*

- Construct a lookup table of all intermediate results:
  $X = E_{K1}(P)$ for all possible keys K_1.
- Decrypt C for all possible keys K_2: $Y = D_{K2}(C)$.
- The key we want is when X = Y.
- Look up each value of Y in the table of X's.
- This takes about twice the effort needed to break the standard single-key DES.
- It is NOT equivalent to using a 56x2 = 112-bit key.

Triple DES

- Three Key Triple DES
  - $C = E_{K3} (D_{K2}(E_{K1}(P)))$
  - No known weaknesses.
  - Decryption with the second key is used so that if the same key is used three times then this is equivalent to single DES.
  - There are many legacy documents encrypted with original DES.

- There are 3 options:
  - K_1, K_2, K_3 are all different: 168-bit key
  - $K_1 = K_3$, K_2 different: 112-bit key avoiding meet in the middle.
  - $K_1 = K_2 = K_3$: 56-bit key, equivalent to single DES.

Summary

- DES was an implementation of the Feistel scheme
  - The chosen key length of 56 bits was too short
  - It used fairly random lookup tables in each round.
  - All the operations were fast and required a small amount of computation.
  - The short key length led to it being broken about 20 years after adoption.
  - The triple-DES version is secure and still in use in major applications

Quizzes

1. How are encryption and decryption performed with a Feistel cipher? Prove that decryption undoes encryption.

2. Describe briefly how the DES algorithm implements the Feistel scheme, mentioning, in particular, the role of the key length and also the so-called S-Boxes. You do not need to provide details of the actual S-Boxes, rather describe in general terms their role in the algorithm.

3. Imagine you have the task of designing a hardware encryption device based on DES chips. Your company anticipates that a well-funded organisation will make a serious attempt to break the data encrypted by your device. How would you choose to employ the DES chips?